

Syntax

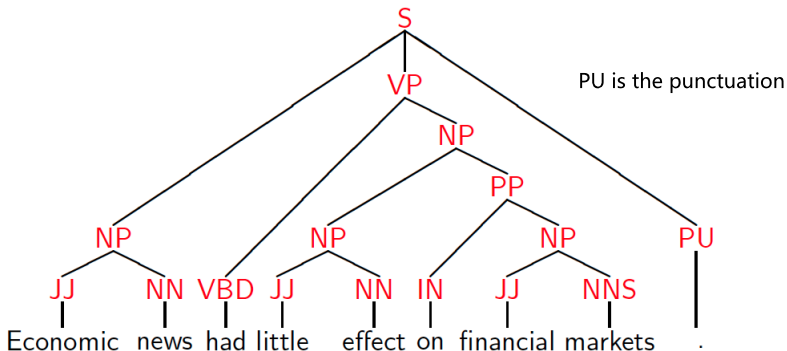
Dependency Graphs

Till now, we studied Constituency Parsing that used Context Free Grammars to capture word groups

**Now we will study another form of parsing - Dependency Parsing
- that tries to infer the relationship between words in a given text**

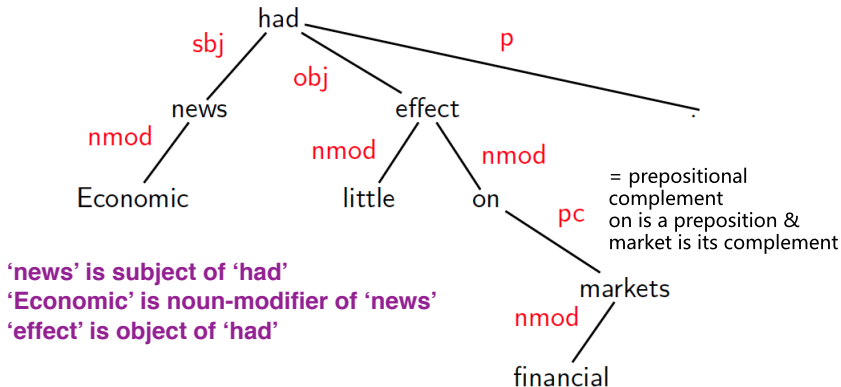
Phrase Structure

A tree structure where the nodes are different Parts-of-speech (JJ, NN, VBD, ...) and phrase-types (NP, VP, PP, ...)



Dependency Structure Representation

A different tree structure where the nodes are the words themselves. The edges (to be considered as directed) denote the relations between the words, e.g., subject, object, ...



In Dependency Parsing every sentence has a root word, here the root word is "had" since it's the main verb of the sentence, everything else depends on main verb directly or indirectly.

Subject Relation : news -> had (subject relation) i.e. news is the subject of "had"

Economic -> news (nmod) Economic modifies "news" hence its a Noun Modifier (nmod)

Dependency Structure

H→D is a construction C

letter → a

Verb → subject

Verb → object

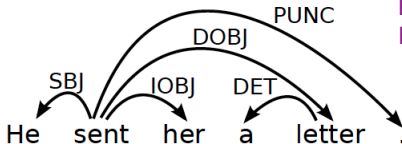
**English mostly
follows
subject-verb-object**

Head → Dependent, in dependency graph, one word is the head (governor) & other word is the dependent

SBJ: subject

IOBJ: indirect object

DOBJ: direct object



Direct Obj is the thing being sent, and IOBJ is the recipient.

- Connects the words in a sentence by putting arrows between the words.
- Arrows show relations between the words and are typed by some grammatical relations.
- Arrows connect a head (governor, superior, regent) with a dependent (modifier, inferior, subordinate).
- Usually dependencies form a tree.

Dependency Graphs

- A dependency structure can be defined as a directed graph G , consisting of
 - ▶ a set V of nodes,
 - ▶ a set A of arcs (edges),
- Labeled graphs:
 - ▶ Nodes in V are labeled with word forms (and annotation).
 - ▶ Arcs in A are labeled with dependency types.

Annotations like POS tags can also be made nodes

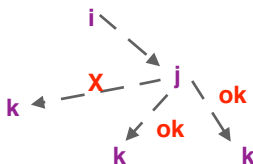
Dependency types: SBJ, OBJ, ...

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- Labeled graphs:
 - ▶ Nodes in V are labeled with word forms (and annotation).
 - ▶ Arcs in A are labeled with dependency types.
- Notational convention:
 - ▶ Arc (w_i, d, w_j) links head w_i to dependent w_j with label d
 - ▶ $w_i \xrightarrow{d} w_j \Leftrightarrow (w_i, d, w_j) \in A$
 - ▶ $i \rightarrow j \equiv (i, j) \in A$
 - ▶ $i \rightarrow^* j \equiv i = j \vee \exists k : i \rightarrow k, k \rightarrow^* j$ **There is a path from i to j**

Formal conditions on Dependency Graphs

- G is connected:
 - ▶ For every node i there is a node j such that $i \rightarrow j$ or $j \rightarrow i$.
- G is acyclic:
 - ▶ if $i \rightarrow j$ then not $j \rightarrow^* i$.
- G obeys the single head constraint:
 - ▶ if $i \rightarrow j$ then not $k \rightarrow j$, for any $k \neq i$.
- G is projective:
 - ▶ if $i \rightarrow j$ then $j \rightarrow^* k$, for any k such that both j and k lie on the same side of i .

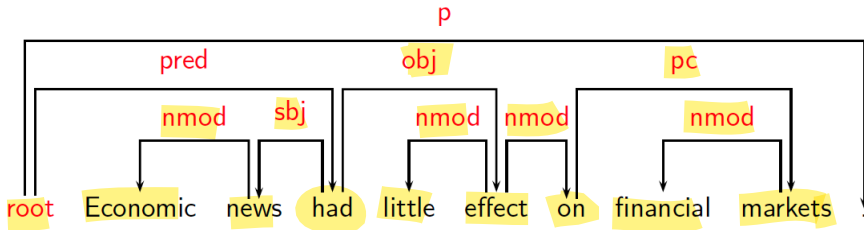


Projectivity is
language-dependent

Formal Conditions: Basic Intuitions

Connectedness, Acyclicity and Single-Head

- **Connectedness:** Syntactic structure is complete.
- **Acyclicity:** Syntactic structure is hierarchical.
- **Single-Head:** Every word has at most one syntactic head.
- **Projectivity:** No crossing of dependencies.



Dependency Parsing

Dependency Parsing

- **Input:** Sentence $x = w_1, \dots, w_n$
- **Output:** Dependency graph G

Parsing Methods

- Deterministic Parsing
- Maximum Spanning Tree Based
- Constraint Propagation Based

The first two methods rely on labeled training data (data-driven parsing). The third method does not need any labeled data, but needs to know some constraints that the grammar follows.